

iSAQB Module: Formal Methods - Logic

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Logic

- Propositional Logic
- Predicate (First-Order) Logic
- Temporal Logic

Moreover:

- Model Theory vs.
- Proof Theory

Propositional Logic

Propositional formulae $F, G, \dots \in \mathcal{D}$

- *atomic formulae* $A, B, C, \dots \in \mathcal{A} = \{A_1, A_2, A_3, \dots\}$
- *constant literals* $\top, \perp$
- *negation* $\neg A$
- *literals* $L \in \mathcal{L} = \{\top, \perp, A_1, A_2, A_3, \dots, \neg A_1, \neg A_2, \neg A_3, \dots\}$
- *disjunction* $(F \vee G)$
- *conjunction* $(F \wedge G)$

Notation

- *implication* $(F \Rightarrow G) \equiv (\neg F \vee G)$
- *equivalence* $(F \Leftrightarrow G) \equiv ((F \Rightarrow G) \wedge (G \Rightarrow F))$

Exercise: General Negation

Negation is restricted to literals in this language. Why do you think that is? How could general negation be recovered?

Semantics

- truth values $\mathcal{T} = \{0, 1\}$
- assignment $A : \mathcal{A} \rightarrow \mathcal{T}$

Extend assignment to *semantics/model* $A' : \mathcal{D} \rightarrow \mathcal{T}$.

1. $A'(\top) = 1, A'(\perp) = 0$
2. $A_i \in \mathcal{D} : A'(A_i) = A(A_i), A'(\neg A_i) = 1 - A(A_i)$
3. $A'(F \wedge G) = \begin{cases} 1 & A'(F) = 1 \text{ and } A'(G) = 1 \\ 0 & \text{otherwise} \end{cases}$
4. $A'(F \vee G) = \begin{cases} 1 & A'(F) = 1 \text{ or } A'(G) = 1 \\ 0 & \text{otherwise} \end{cases}$

Validity

A formula F is *valid* if $A'(F) = 1$ in every assignment A / model A' .

Exercise: Semantics of Propositional Logic

Calculate the semantics of $\neg((A \wedge B) \vee C)$ for the assignment $A(A) = 1, A(B) = 1, A(C) = 0$!

Semantics Notation & Terminology

- $A \models F \equiv A'(F) = 1$: A is a *model* for F
- model exists: F is *satisfiable*
- no model exists: F is *unsatisfiable*
- all assignments are models: F is a *tautology*
- $A \models F$ iff $A \models G$ for all A : F and G are *equivalent*

Inference Calculus

- *axioms*
- *inference rules*
- *derivability/provability*

Sequent Calculus

Multi-sets (*sequences*) of formulae $\Gamma, \Pi, \Delta, \dots$

Axioms: $\top, \Gamma$ $\neg A_i, A_i, \Gamma$

Inference rules:

$$\frac{F, G, \Gamma}{F \vee G, \Gamma}$$

$$\frac{F, \Gamma \quad G, \Gamma}{F \wedge G, \Gamma}$$

Derivability

Formula is derivable $\vdash F$ iff a tree labelled with sequences exists where:

- root is the formula
- leaves are axioms
- each internal node is an instance of an inference rule

Properties of a calculus

- *consistency* contradictions are not derivable
- *soundness* only tautologies are derivable
- *completeness* all tautologies are derivable
- *decidable* it is decidable whether a formula is derivable or not

Theorem: The sequent calculus all of the above.

Exercise: Sequent Calculus Proofs

- Prove $(A_1 \wedge A_2) \vee (\neg A_1 \vee \neg A_2)$ in the sequent calculus!
- What is a sensible strategy to construct a proof for a given formula?

Classical vs. Intuitionistic Logic

In the Sequent Calculus, $F \vee \neg F$ (*LEM*) and $F \Leftrightarrow \neg\neg F$ (double negation) are derivable for any F .

This enables *proof by contradiction*.

This is problematic in some settings, as proofs by contradiction are not *constructive*.

Constructive Proofs

Theorem There are irrational numbers a and b such that a^b is rational.

Exercise: What would you expect from a proof of this theorem?

Non-Constructive Proof

Theorem There are irrational numbers a and b such that a^b is rational.

Exercise: What are two numbers a and b such that a^b is rational?

Non-Constructive Proof

Consider $c = \sqrt{2}$. c^c is either rational, or not.

1. Assume c^c is rational. Then we can choose $a = b = c$ and we're done.
2. Assume c^c is irrational. Then we can choose $a = c^c = \sqrt{2}^{\sqrt{2}}$ and $b = c$. Both are irrational.

$$a^b = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^2 = 2$$

Intuitionistic logic would not admit this proof.

Predicate Logic

- *variables* $u, v, w, x, y, z, \dots \in \mathcal{V} = \{x_1, x_2, x_3, \dots\}$
- *function symbols* $f, g, h, \dots \in \mathcal{F} = \{f_1, f_2, f_3, \dots\}$
- each function symbol has *arity* $k \in \mathcal{N}$

Constants $a, b, c, \dots$ are function symbols with arity 0.

Terms $a, b, c, \dots, t_1, \dots, t_n$:

1. Each variable is a term.
2. $f(t_1, \dots, t_k)$ if f has arity k

Formulae

- *predicate symbols* $P, Q, R, \dots \in \mathcal{P} = \{P_1, P_2, P_3, \dots\}$
- each predicate symbol has *arity* $k \in \mathbb{N}$

Formulae $F, G, \dots \in \mathcal{D}$:

1. *constant literals* $\top, \perp$
2. *atomic formulae* $P(t_1, \dots, t_k)$ if P predicate symbol with arity k
3. $\neg F$
4. $(F \vee G)$
5. $(F \wedge G)$
6. $(\exists x.F)$ ($\exists$ *existential quantifier*)
7. $(\forall x.F)$ ($\forall$ *universal quantifier*)

Free, Bound Variables

Each *occurrence* of a variable is either *bound* or *free*.

An occurrence of the variable x in the formula F is *bound* if x occurs within a subformula of F of the form $\exists x.F$ or $\forall x.F$.

An occurrence of the variable x in the formula F is *free* if it is not bound.

Formula without occurrence of a free variable: *closed*.

Exercise: Free and Bound Variables

Write a formal definition of free and bound variables!

Semantics

Structure S:

- non-empty set U *universe*
- *function assignment* mapping each k -ary function symbol $f \in \mathcal{F}$ to a k -ary function

$$\bar{f} : U \times \dots \times U \rightarrow U$$

- *predicate assignment* mapping each k -ary predicate $P \in \mathcal{P}$ to a k -ary predicate

$$\bar{P} \in U \times \dots \times U$$

Variable assignment $\beta : \mathcal{V} \rightarrow U$

Function, predicate, and variable assignments form an *interpretation*.

Semantics

For a given structure S and variable assignment β , the *model* $A_{S\beta} : \mathcal{D} \rightarrow \mathcal{T}$ of a formula is as follows:

1. $A_{S\beta}(\top) = 1, A_{S\beta}(\perp) = 0$
2. $x \in \mathcal{V} : A_{S\beta}(x) = \beta(x)$
3. 0-ary $a \in \mathcal{F} : A_{S\beta}(a) = \bar{a}$
4. k -ary $f \in \mathcal{F} : A_{S\beta}(f(t_1, \dots, t_k)) = \bar{f}(A_{S\beta}(t_1), \dots, A_{S\beta}(t_k))$
5. k -ary $P \in \mathcal{P} : A_{S\beta}(P(t_1, \dots, t_k)) = \begin{cases} 1 & (A_{S\beta}(t_1), \dots, A_{S\beta}(t_k)) \in \bar{P} \\ 0 & \text{otherwise} \end{cases}$

Semantics

$$6. A_{S\beta}(\neg F) = \begin{cases} 1 & A_{S\beta}(F) = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$7. A_{S\beta}(F \wedge G) = \begin{cases} 1 & A_{S\beta}(F) = 1 \text{ and } A_{S\beta}(G) = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$8. A_{S\beta}(F \vee G) = \begin{cases} 1 & A_{S\beta}(F) = 1 \text{ or } A_{S\beta}(G) = 1 \\ 0 & \text{otherwise} \end{cases}$$

Semantics

$$\begin{aligned} 9. A_{S\beta}(\exists x_i.F) &= \begin{cases} 0 & A_{S\beta'}(F) = 0 \\ & \text{for each variable assignment } \beta' \\ & \text{with } \beta'(x_j) = \beta(x_j) \text{ for } j \neq i \\ 1 & \text{otherwise} \end{cases} \\ 10. A_{S\beta}(\forall x_i.F) &= \begin{cases} 1 & A_{S\beta'}(F) = 1 \\ & \text{for each variable assignment } \beta' \\ & \text{with } \beta'(x_j) = \beta(x_j) \text{ for } j \neq i \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Terminology

- *satisfiable, unsatisfiable, tautology* all as with propositional logic

Sequent Calculus

Propositional calculus +

$$\frac{F[x \mapsto b], \exists x.F, \Gamma}{\exists x.F, \Gamma}$$

$$\frac{F[x \mapsto y], \Gamma}{\forall x.F, \Gamma} \text{ where } y \notin \text{free}(\forall x.F) \cup \text{free}(\Gamma)$$

Exercise: Correctness of Lane-Change Assistant

Write a criterion for the correctness of a lane-change assistant in predicate logic.

A lane-change assistant first looks for a gap in traffic either on the left or right of the car. If it finds a gap, it will change to the lane of the gap.

Linear Temporal Logic

Formulae of LTL $\phi, \rho, \dots \in \mathcal{L}$:

- $\top, \perp$
- *atomic propositions* $A, B, C, \dots \in \mathcal{A} = \{A_1, A_2, A_3, \dots\}$
- *negation* $\neg\phi$
- *disjunction* $(\phi \vee \rho)$
- *conjunction* $(\phi \wedge \rho)$
- *next* $(\mathbf{X}\phi)$
- *until* $(\phi \mathbf{U} \rho)$

LTL Semantics

- *valuation* $\sigma \in \mathcal{P}(\mathcal{A})$ true propositions
- *computation* $\sigma = \sigma_0\sigma_1\sigma_2 \dots \in \mathcal{P}(\mathcal{A})^\omega$
infinite sequence of valuations
- $\sigma^i = \sigma_i\sigma_{i+1}\sigma_{i+2} \dots$

LTL Satisfaction Relation

$\sigma \models \phi$:

- $\sigma \models \top$
- $\sigma \models A$ iff $A \in \sigma_0$
- $\sigma \models \neg A$ iff $\sigma \not\models A$
- $\sigma \models \phi \vee \rho$ iff $\sigma \models \phi$ or $\sigma \models \rho$
- $\sigma \models \phi \wedge \rho$ iff $\sigma \models \phi$ and $\sigma \models \rho$
- $\sigma \models \mathbf{X}\phi$ iff $\sigma^1 \models \phi$
- $\sigma \models \phi \mathbf{U} \rho$ iff there is $k \geq 0$ with
 $\sigma^k \models \rho$ and
 $\sigma^i \models \phi$ for all $0 \leq i < k$

LTL Syntactic Sugar

- *eventually* $(\mathbf{F}\phi) = \top \mathbf{U} \phi$
 $\sigma \models \mathbf{F}\phi$ iff there is $k \geq 0$ with $\sigma^k \models \phi$
- *always* $(\mathbf{G}\phi) = \neg(\mathbf{F}\neg\phi)$
 $\sigma \models \mathbf{G}\phi$ iff $\sigma^k \models \phi$ for every $k \geq 0$

Exercise: LTL Tutor

Do a few exercises in the **LTL Tutor**!

<https://ltl-tutor.xyz/>

Exercise: Vending Machine State Diagram

Consider a vending machine that:

- accepts a coin
- then allows selecting tea or coffee
- after dispensing the drink, goes back to the beginning

Draw a state diagram for the vending machine, and formulate LTL properties that a vending machine should fulfill.

Kripke Structure

A *Kripke structure* is a tuple $(W, \leadsto, w, I)$ where:

- W set of *states*
- $\leadsto \in W \times W$ *transition relation*
- $w : W \rightarrow \mathcal{P}(\mathcal{A})$ maps states to valuations
- $I \subseteq W$ *initial states*

Computation Structure

A Kripke structure is a *computation structure* if:

- W is finite
- every $s \in W$ has a successor t with $s \rightsquigarrow t$.

A *computation path* is an infinite sequence $s_0 s_1 s_2 \dots$ with $s_i \in W$ and $s_i \rightsquigarrow s_{i+1}$ for all i .

LTL Semantics / Computation Structure

Given an LTL formula ϕ and a computation structure $K = (W, \simeq, w, I)$, then

$$K \models \phi$$

$$\text{iff } \sigma \models \phi$$

for all σ where

$$\sigma_i = w(s_i) \text{ for some computation path } s_0 s_1 s_2 \dots \text{ in } K$$